



Batterjee Medical College
For Science and Technology

Approach to a case of Dyspnea

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Valvular Heart Disease (VHD)

Learning Objectives

Knowledge & understanding

Describe the clinical features, characteristic auscultatory findings, and complications of each major valvular heart disease.

Describe the types, indications, and thromboembolic risk factors associated with prosthetic heart valves

Skill

Select the appropriate surgical or medical management strategy for a patient with valvular heart disease based on severity, symptoms, and left ventricular function.

Formulate an urgent management plan for a patient with acute, severe valvular regurgitation.

Main Subtopics

- Overview and Classification of Valvular Heart Disease (stenosis vs. regurgitation; anatomical and auscultatory sites)
- Pathophysiologic Effects of Valvular Stenosis and Regurgitation on Cardiac Chambers
 - Aortic Regurgitation
 - Aortic Stenosis
 - Mitral Regurgitation
 - Mitral Valve Prolapse
 - Mitral Valve Stenosis
 - Tricuspid Regurgitation
 - Prosthetic Heart Valves

Case scenario :

A 75-year-old woman is brought to the emergency department 30 minutes after an episode of dizziness and loss of consciousness that lasted 10 seconds. The symptoms occurred while the patient was exercising in an aerobics class. She has a 3-year history of progressive shortness of breath and chest discomfort on exertion. Radial pulses are decreased and delayed bilaterally.

Case problem :

The patient most likely has a murmur starting at which of the following labeled points in the curve shown?

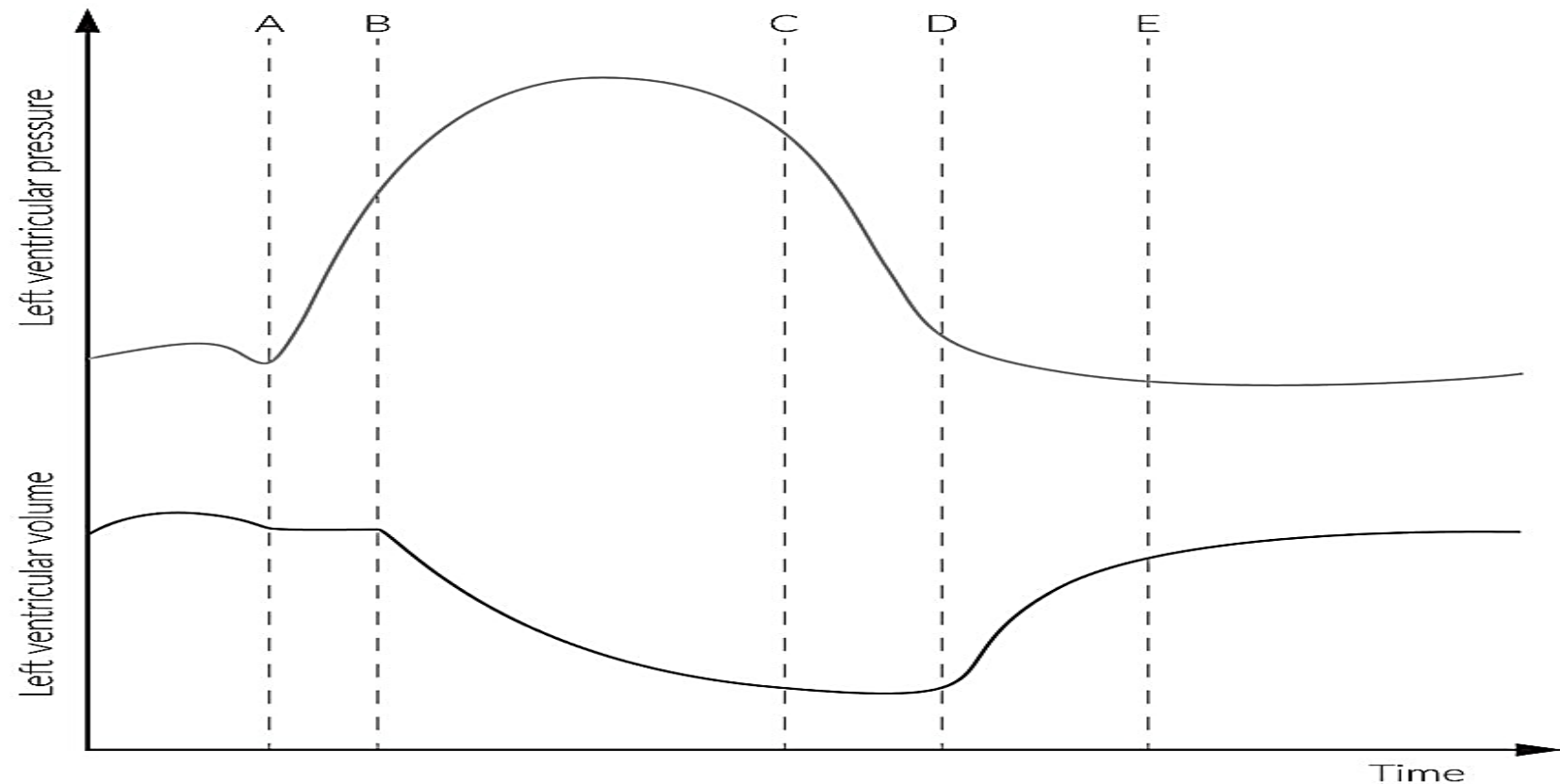
A. A

B. B

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D. D

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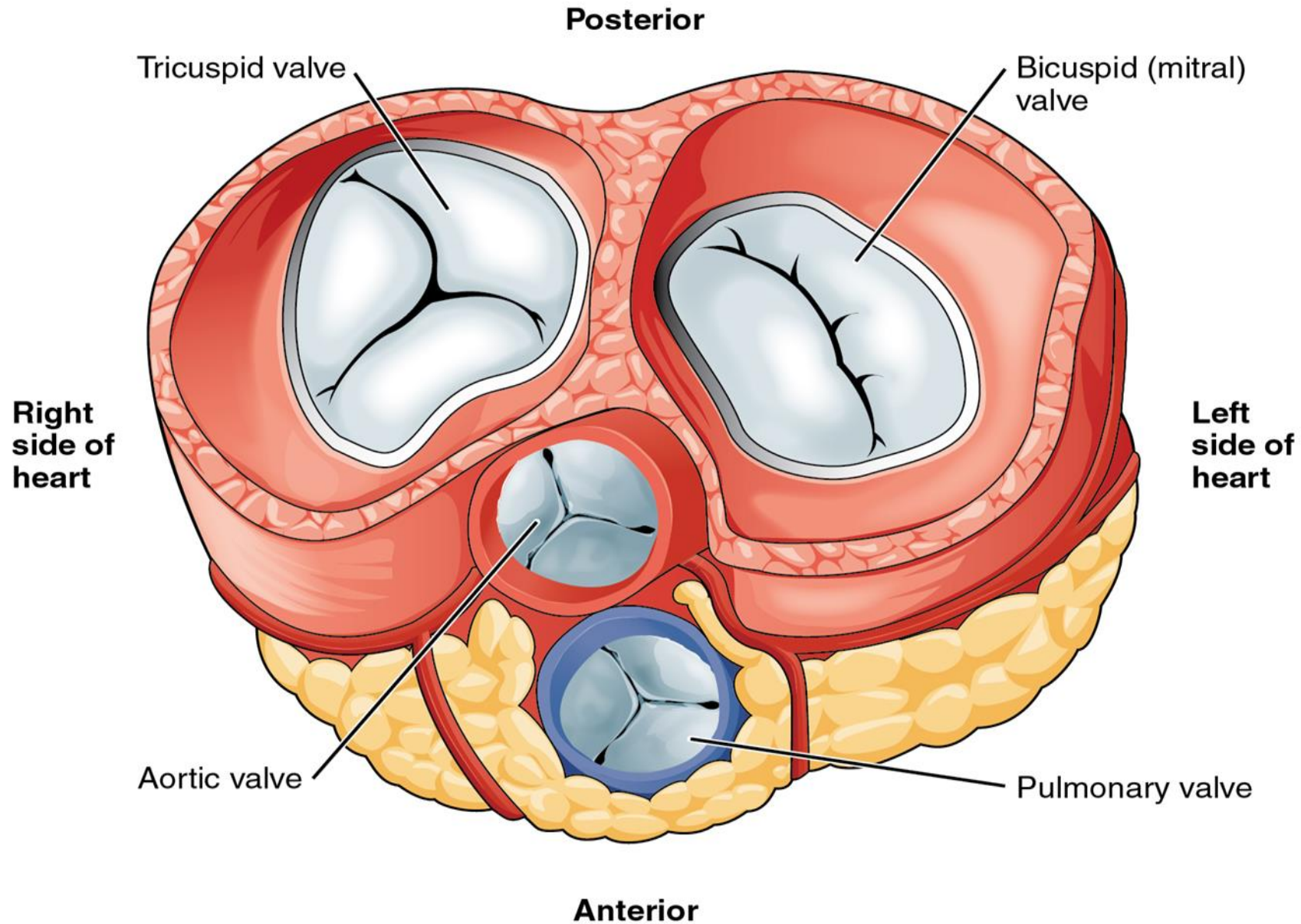


Valvular Heart Disease (VHD)

A group of conditions that affect the heart valves.

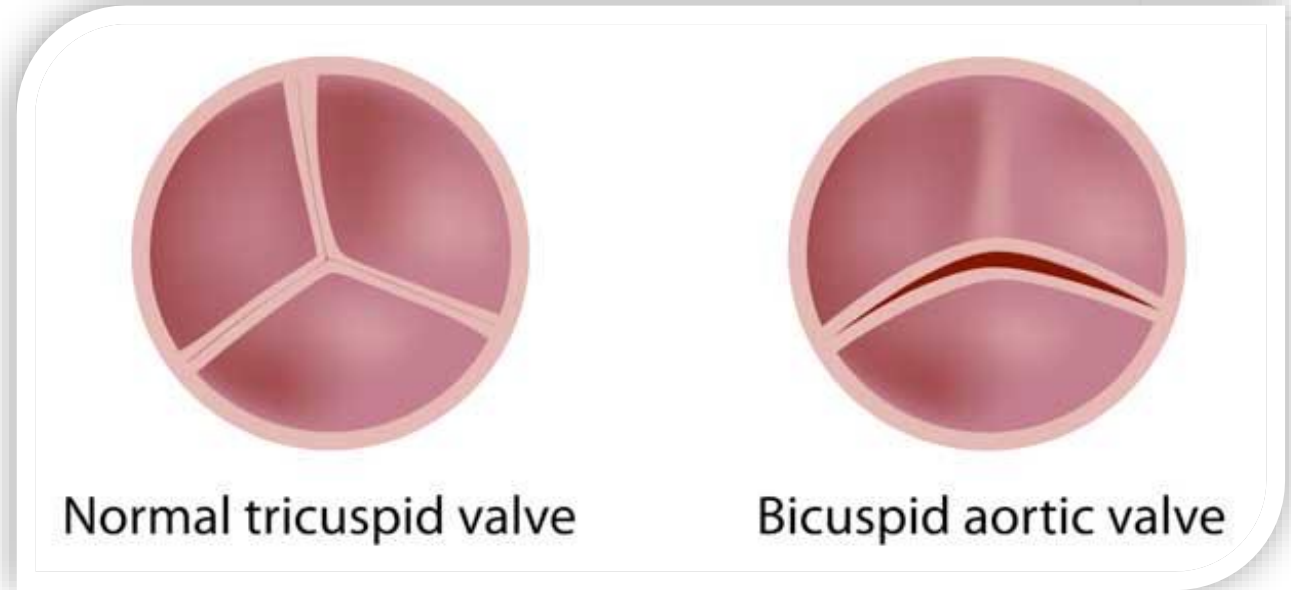
Valvular defects are either **acquired** or **congenital** and manifest as **stenosis** and/or **insufficiency** (regurgitation) of the valves.

Valvular Heart Disease (VHD)



Aortic stenosis

- Most common valve defect in industrialized countries
- Mostly degenerative
- Usually becomes symptomatic after the age of 75
- Aortic stenosis in young people is usually secondary to congenital defects (e.g., bicuspid aortic valve)



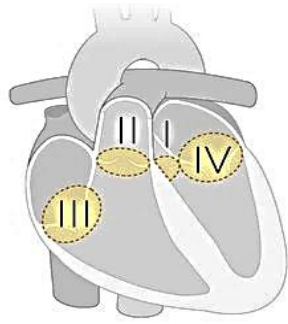
Mitral stenosis

Symptom onset between 20 and 39 years

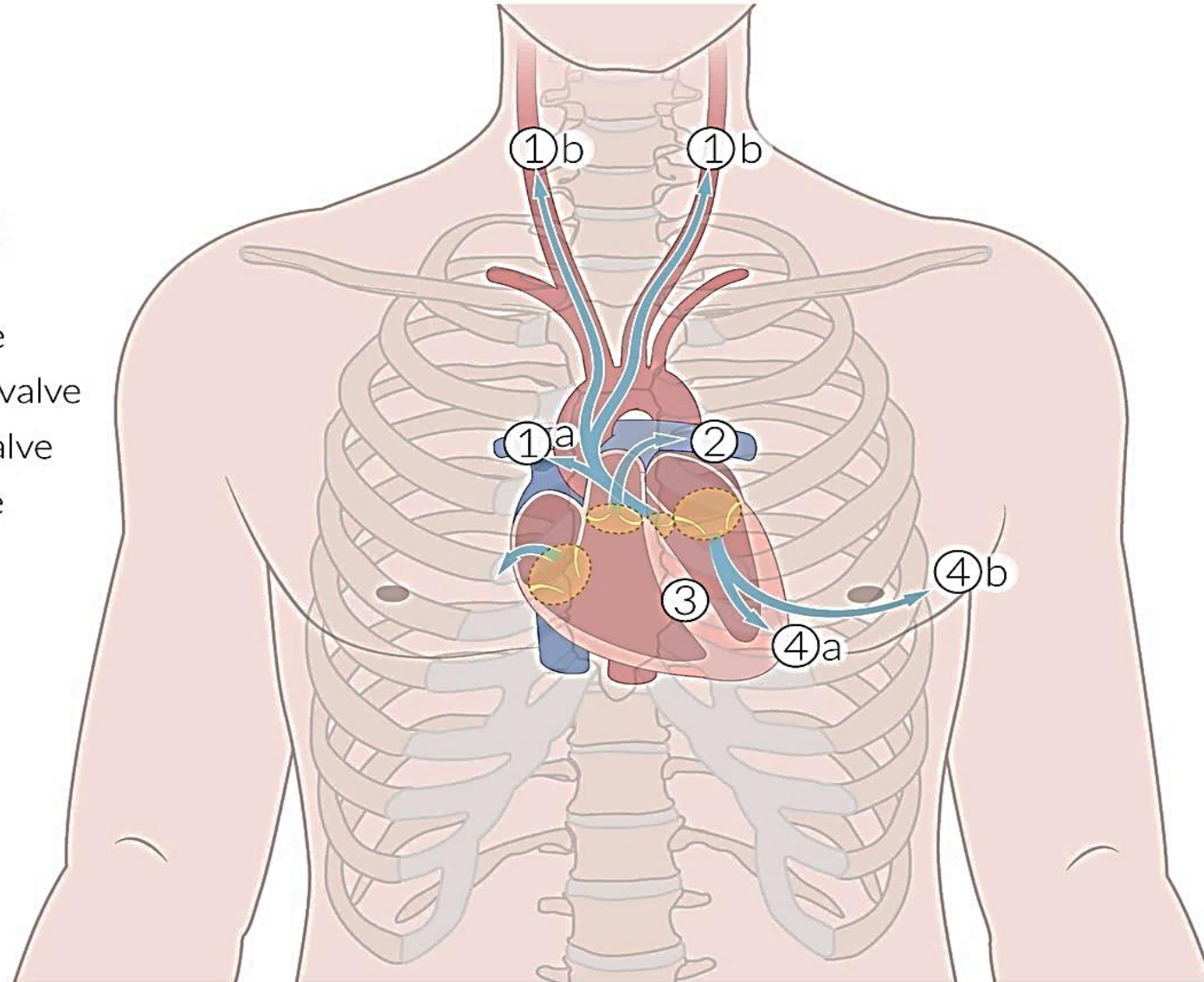
Mitral regurgitation

Second most common valve defect

Anatomical locations and auscultatory sites for heart valves



- I Aortic valve
- II Pulmonary valve
- III Tricuspid valve
- IV Mitral valve



Effect of Valvular heart disease

Valvular stenosis:

- **Effects:** The chamber behind the stenotic valve is subject to greater **stress** and so, **generate more pressure** to force blood through the narrowed opening. This additional workload is compensated by **hypertrophy**, dilatation of the myocardium, and then **heart failure**.

Valvular Regurgitation

- **Effects:** This causes them to **pump the same blood twice** (as the blood comes back into the chamber), the heart **dilates** to accommodate more blood, and ventricular dilation then eventually leads to **heart failure**.



Valvular Heart Disease

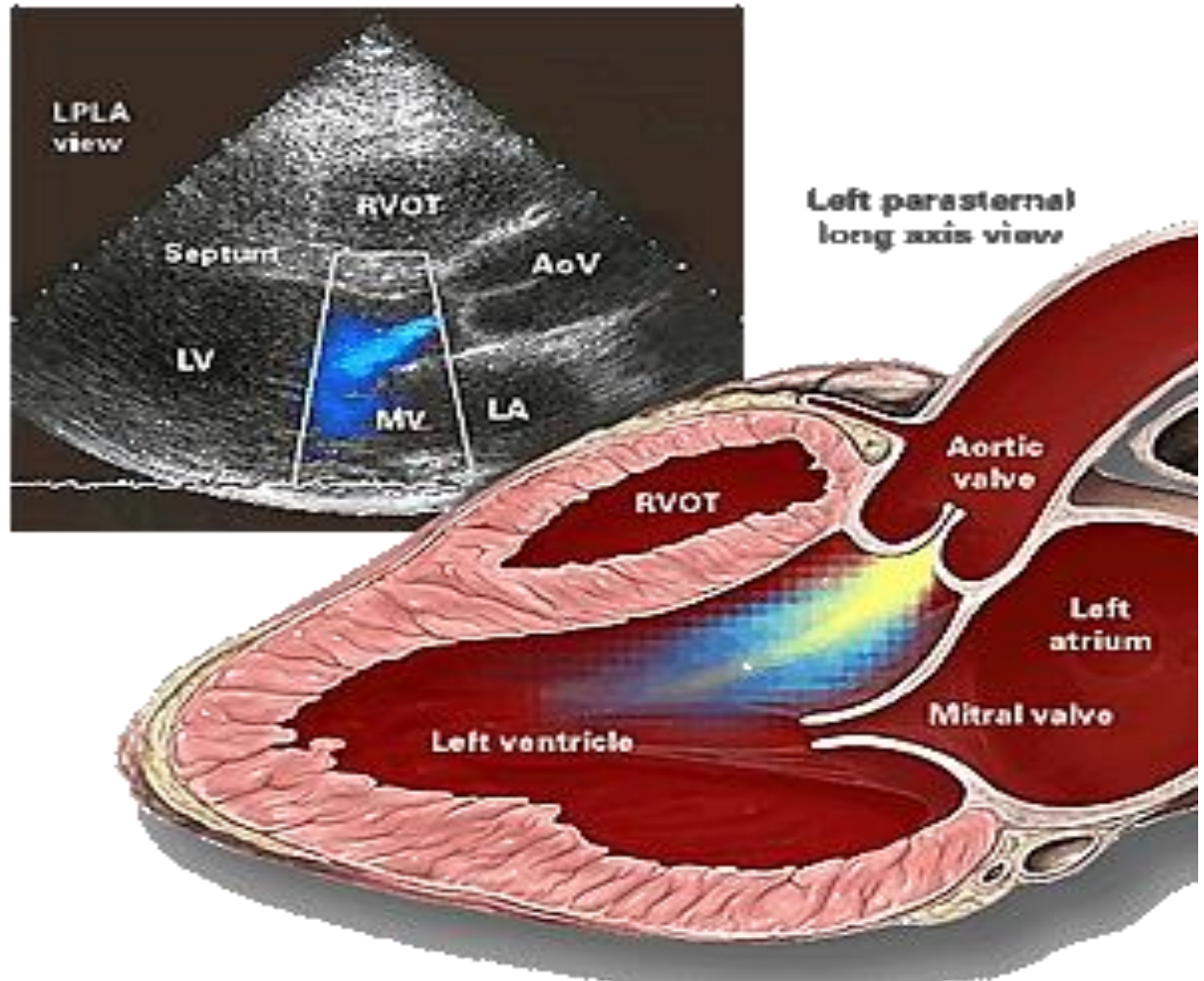
Aortic regurgitation

(Aortic insufficiency)

Aortic regurgitation (Aortic insufficiency)

Acute aortic regurgitation

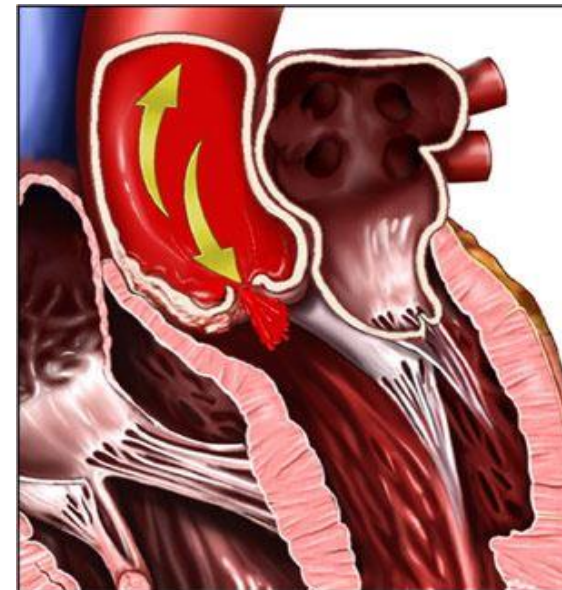
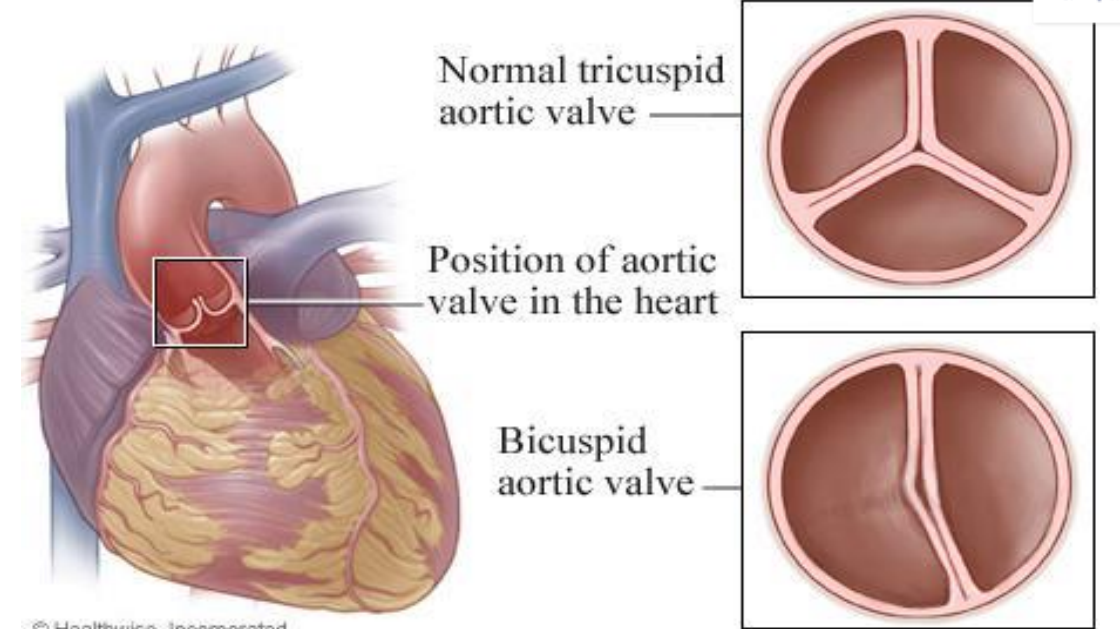
- Infective endocarditis
- Aortic dissection
- Chest trauma



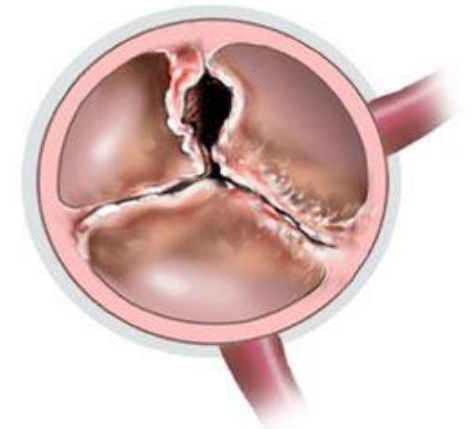
Aortic regurgitation (Aortic insufficiency)

Chronic aortic regurgitation

- Primary valvular defect
- Congenital bicuspid AV
most common cause of AR
in young adults in high-income countries
- Rheumatic heart disease
most common cause of AR
in lower-income countries



View from above closed aortic valve

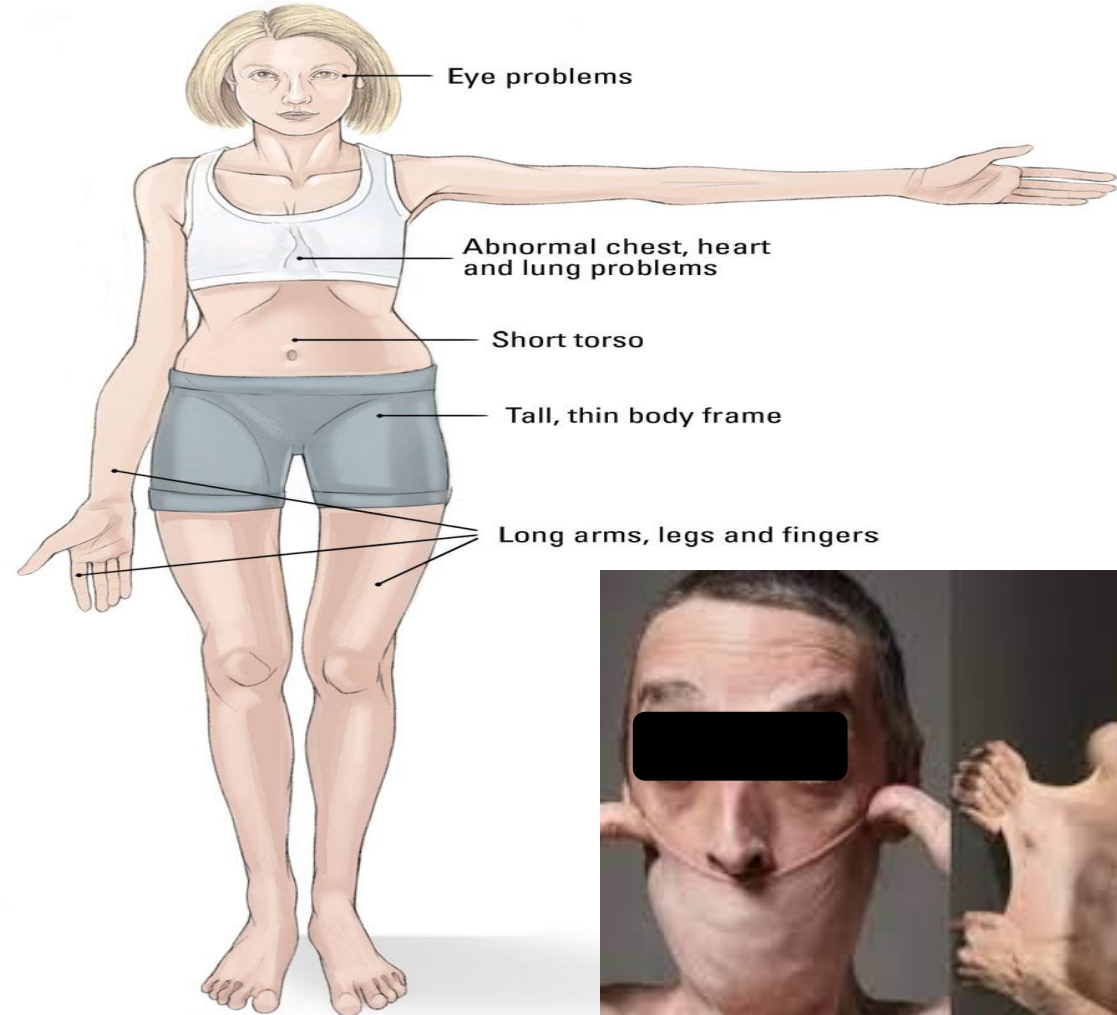


Aortic regurgitation (Aortic insufficiency)

Chronic aortic regurgitation

- Aortic dilatation
- Connective tissue disorder
 - Marfan syndrome
 - Ehlers-Danlos syndrome
- Tertiary syphilis

Marfan syndrome



Ehlers-Danlos syndrome

Acute aortic regurgitation

Signs and symptoms

- Sudden, severe dyspnea
- Rapid cardiac decompensation
- Pulmonary edema

Chronic aortic regurgitation

May be asymptomatic for up to decades despite progressive LV dilation

- Palpitations
- Symptoms of high pulse pressure
 - Water hammer pulse
 - Traube sign: pistol shot-like sounds heard over the femoral artery
 - Quincke sign: visible capillary pulse when pressure is applied to the tip of a fingernail
- Symptoms of left heart failure
 - Exertional dyspnea
 - Angina
 - Orthopnea
 - Easy fatigability
 - Syncope

Clinical features – peripheral manifestations of AR

De Musset's sign: Bobbing of the head with each heartbeat

Becker's sign: Visible pulsation of the retinal arterioles

Muller's sign: Visible pulsations of the uvula

Quincke's sign: Capillary pulsations seen on light compression of the nail bed

Corrigan's pulse: A rapid and forceful distension of the arterial pulse with a quick collapse (**Water hammer pulse**)

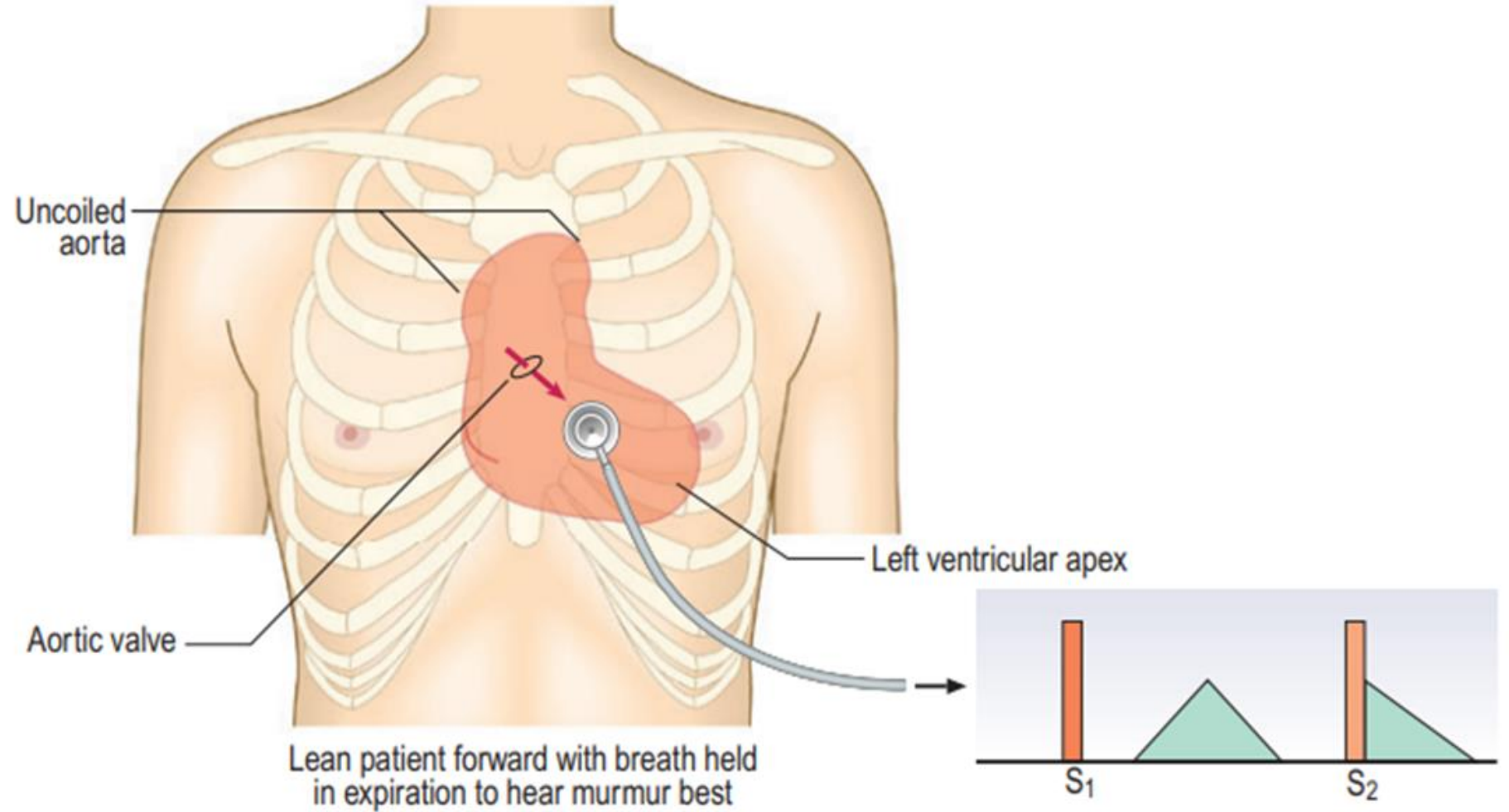
Traube's sign: ("pistol shots")

Duroziez's sign: Gradual pressure over the femoral artery leads to a systolic and diastolic bruit

Hill's sign: Popliteal systolic blood pressure exceeding brachial systolic blood pressure by ≥ 60 mmHg (**most sensitive sign for aortic regurgitation**)

Chronic aortic regurgitation

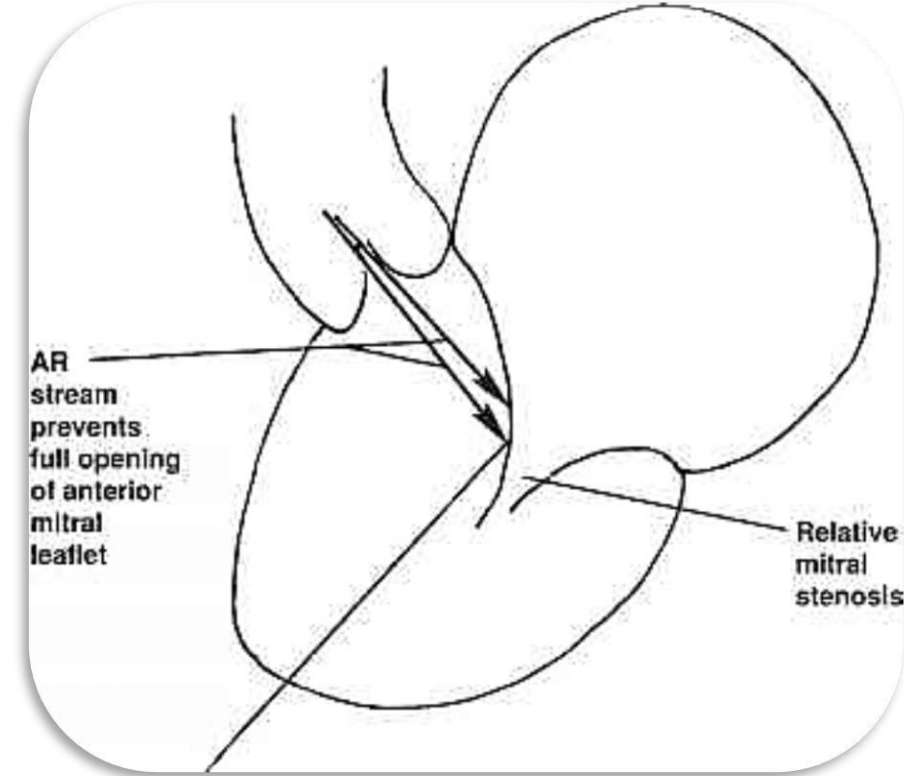
Auscultation



Chronic aortic regurgitation

Auscultation

- S3
- High-pitched, blowing, decrescendo early diastolic murmur
- Austin Flint murmur
- In more severe stages, possibly a harsh, crescendo-decrescendo mid-systolic murmur



Diagnostics of AR

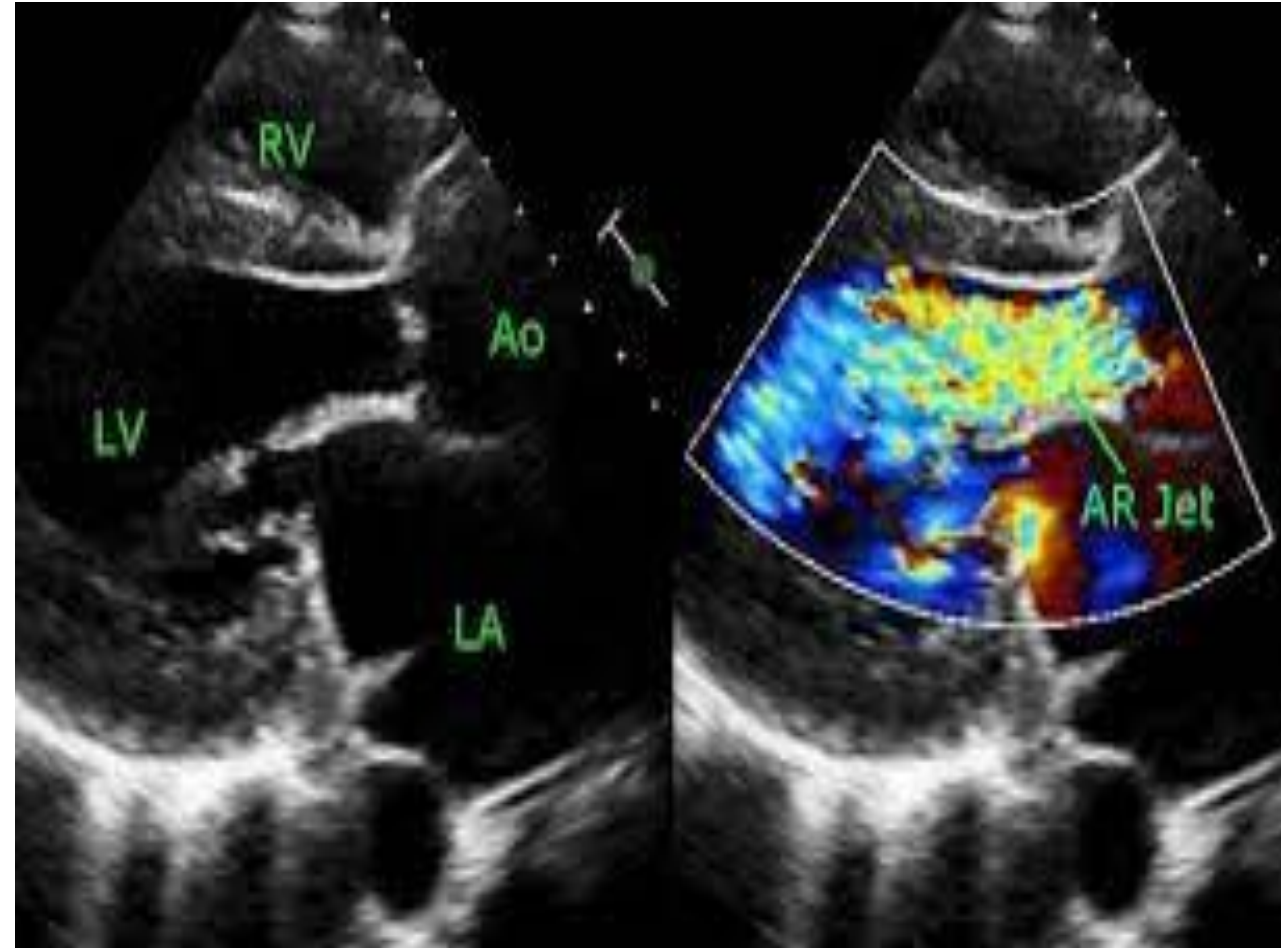
Echocardiography

Indication:

- Assess of AV structure and function
- Cause
- Severity of regurgitation
- Left ventricle and
- Other heart valves

TTE: modality of choice

- Recommended primary test; to confirm diagnosis and determine AS severity



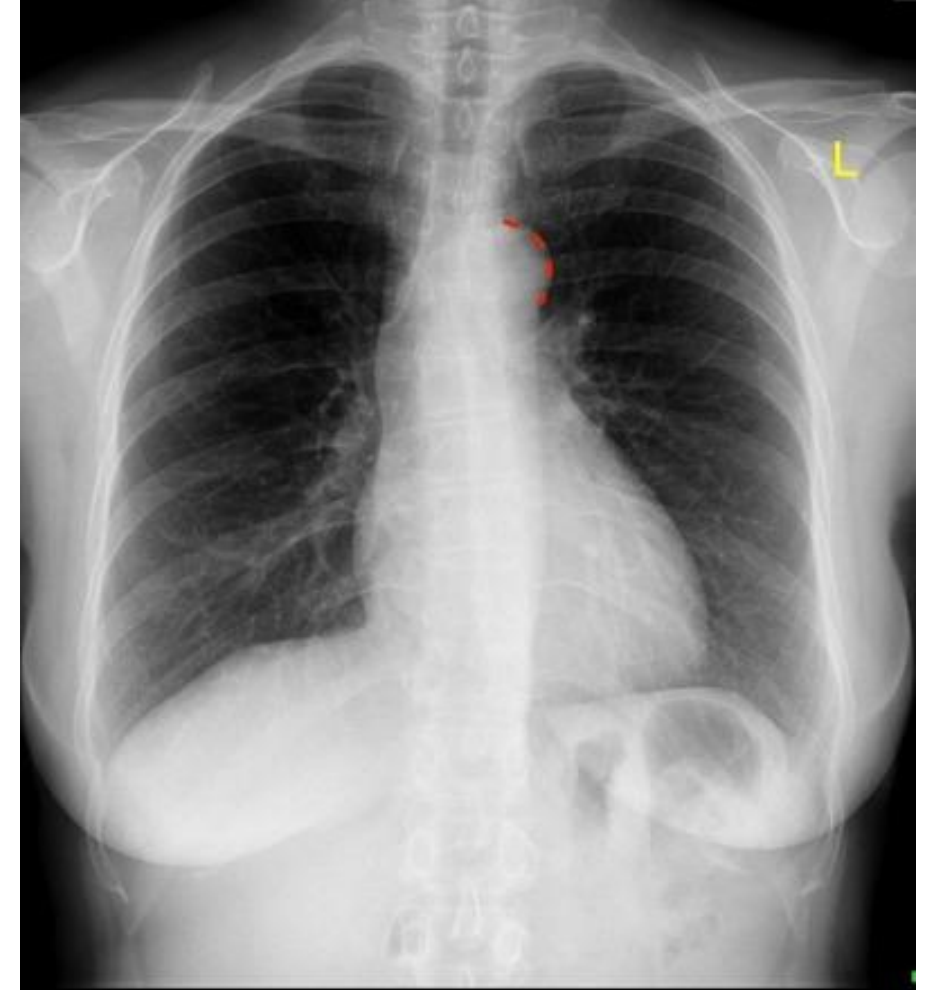
Chest x-ray

used to assess for:

- Pulmonary edema and
- Rule out other causes of acute dyspnea

Finding of chronic AR

- Enlarged cardiac silhouette
- Prominent aortic root/arch



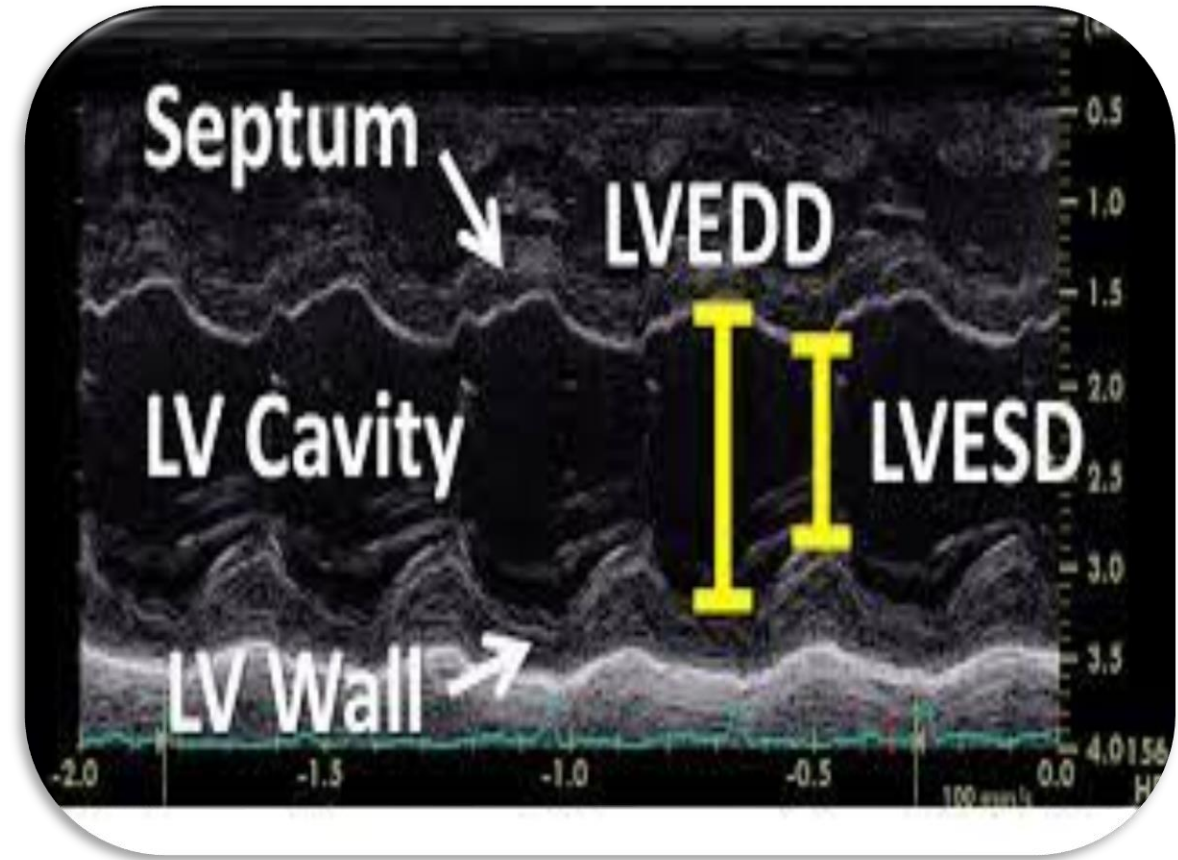
Surgical management

Indications :

- Acute severe AR
- Symptomatic chronic severe AR
- Asymptomatic chronic severe AR with one of the following:
 - Reduced LVEF $\leq 55\%$
 - Consider if LVESD > 50 mm

Medical management

Treatment of heart failure



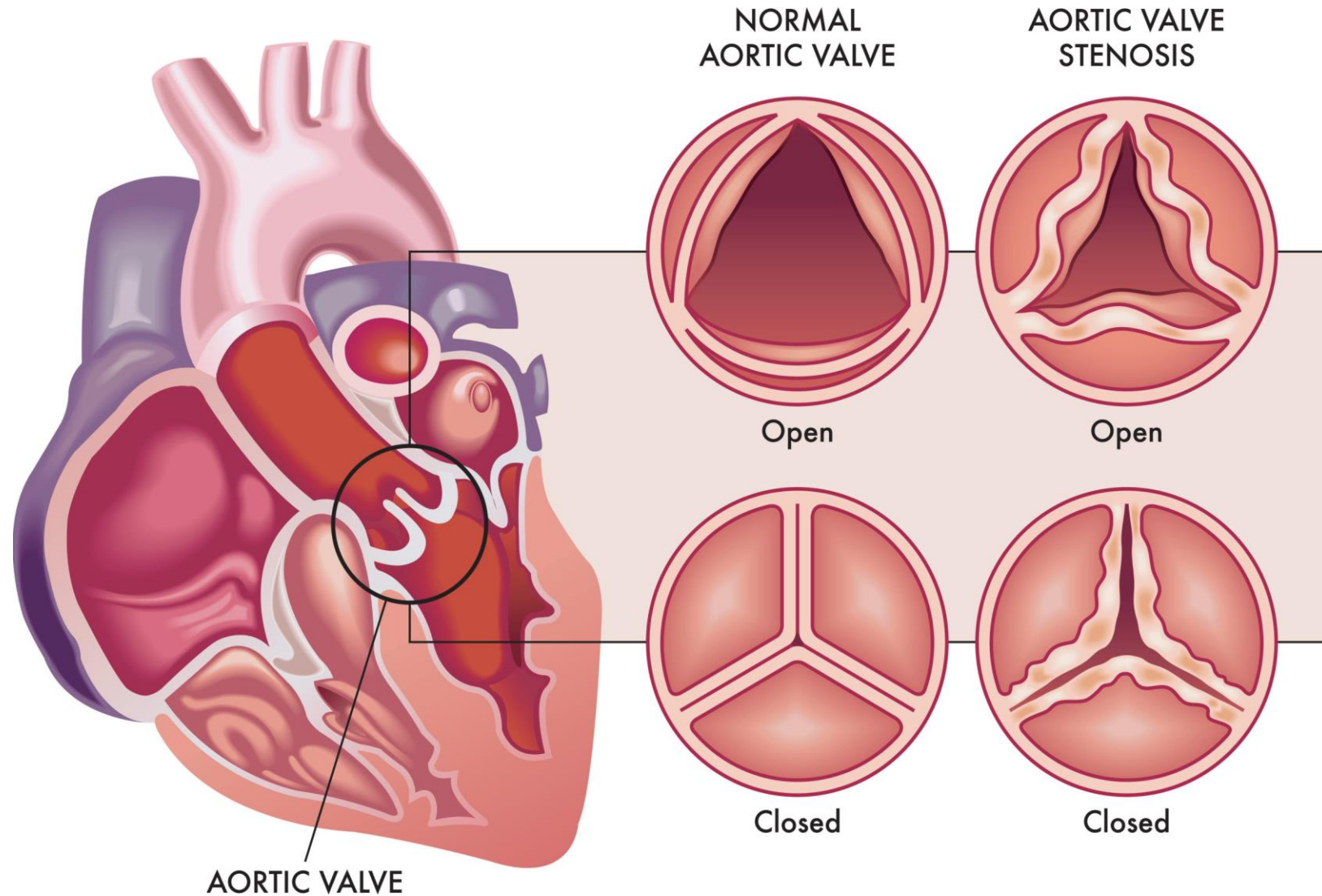


Valvular Heart Disease

Aortic stenosis (AS)

Most common valvular heart disease in industrialized countries

Prevalence increases with age.



Aortic valve sclerosis:

Calcification and fibrosis of aortic valve leaflets

Most common cause of aortic stenosis

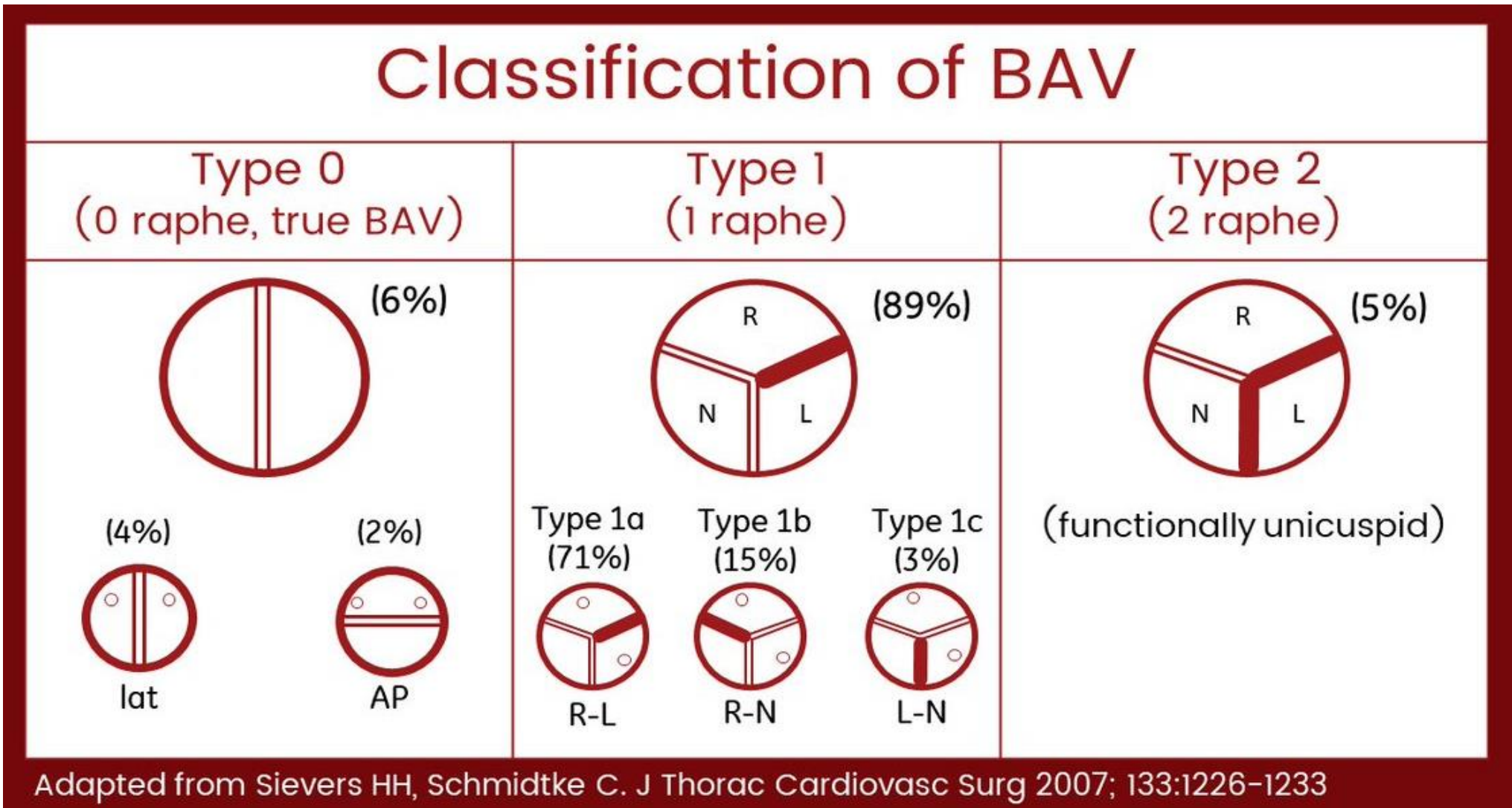
Bicuspid aortic valve (BAV):

Fusion of two of the three aortic-valve leaflets in utero

Patients present with symptoms of aortic stenosis earlier than in regular aortic valve calcification.

Rheumatic fever

Bicuspid aortic valve (BAV):



AS - Clinical features

Aortic stenosis may remain asymptomatic for years

Signs and symptoms(SAD)

Dyspnea

Angina pectoris

Syncope and Dizziness

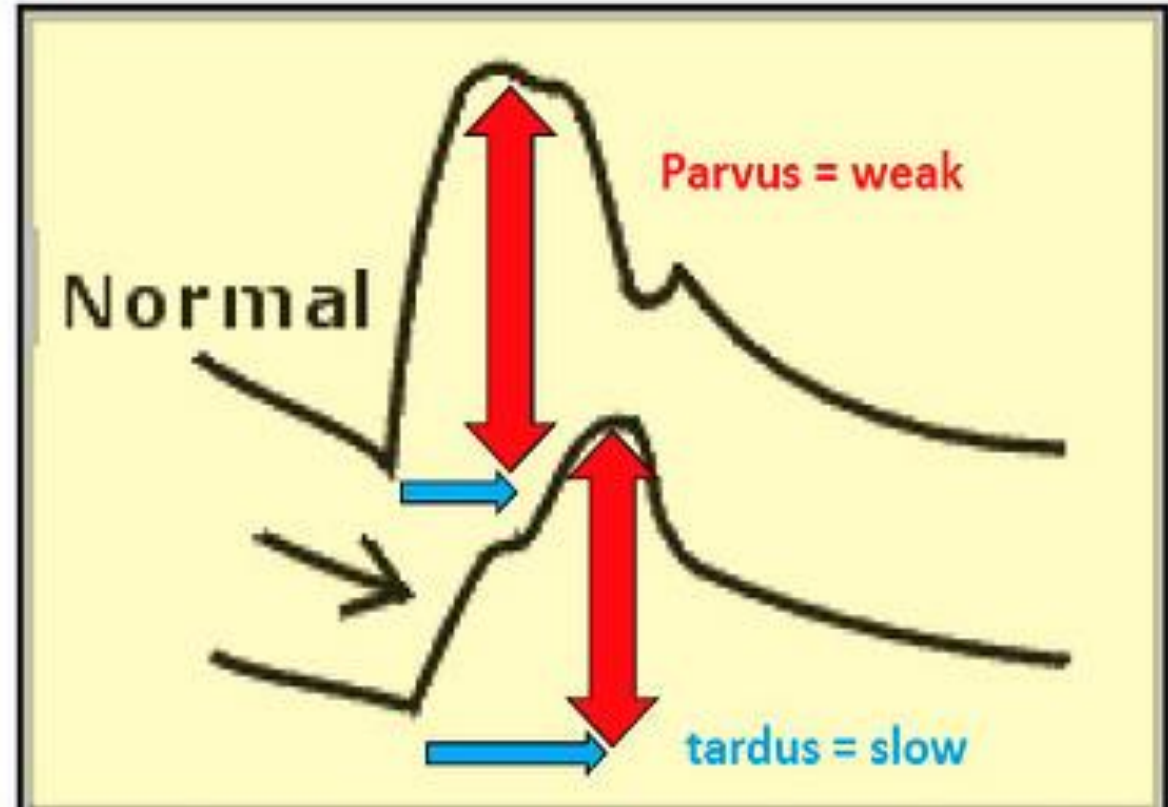
Physical examination

Small blood pressure amplitude
decreased pulse pressure

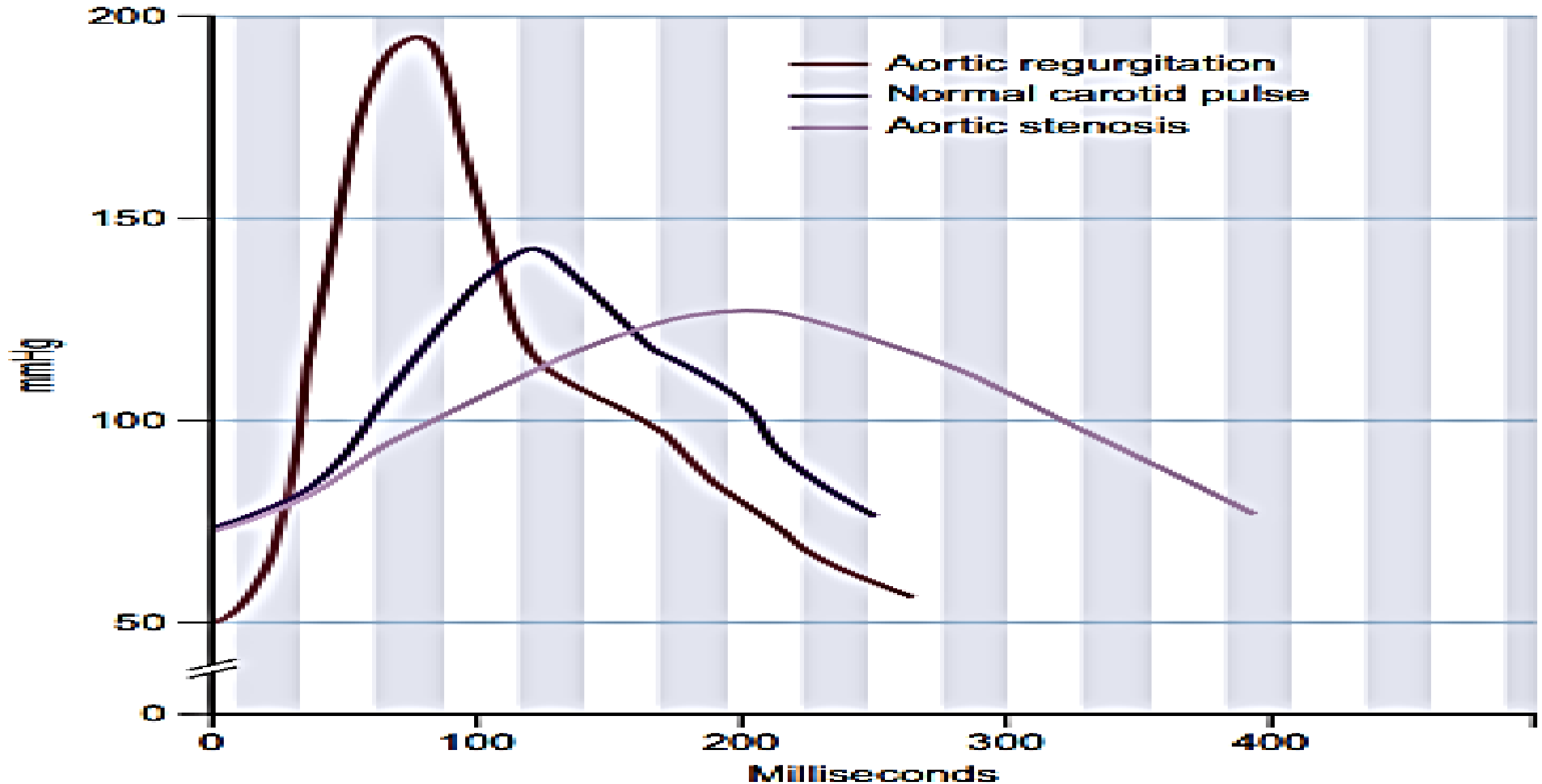
Weak and delayed distal pulse ;
pulsus **parvus** et **tardus**



Pulsus parvus et tardus



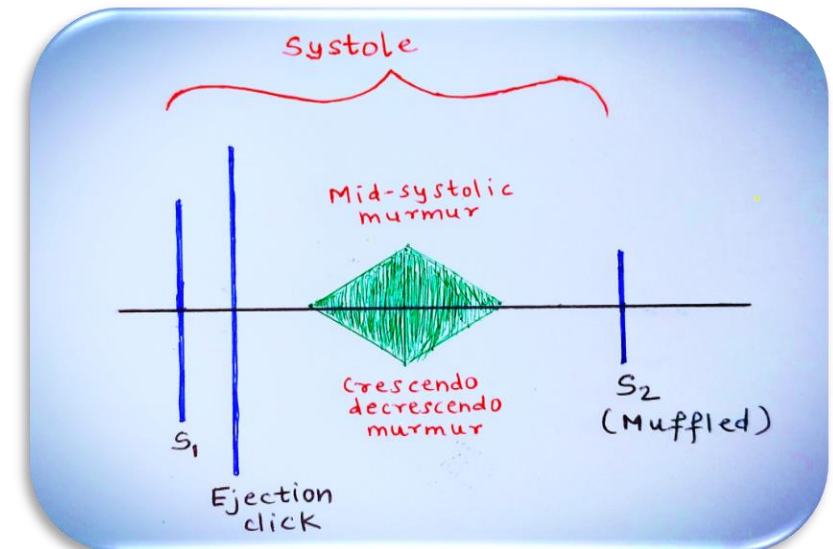
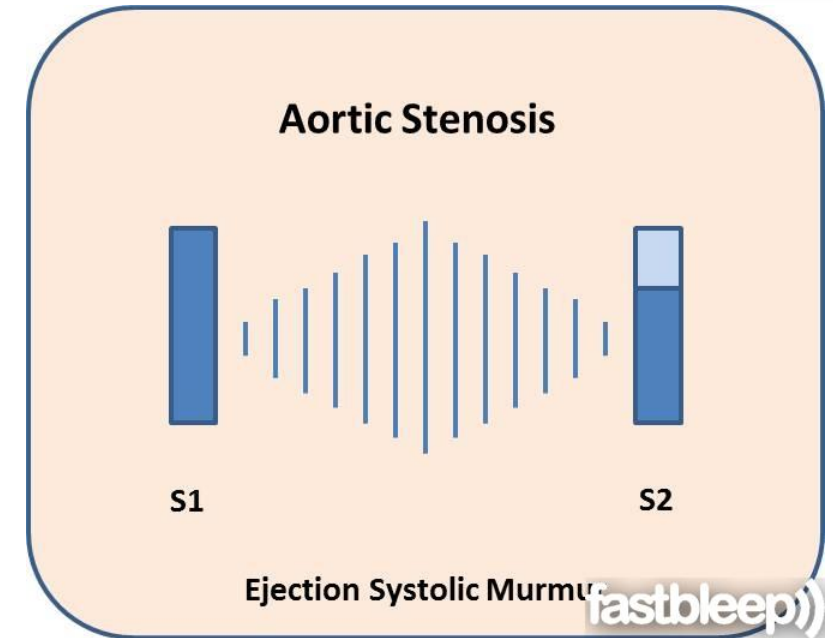
Pulse waveforms



AS - Clinical features

Auscultation

- Murmur harsh crescendo-decrescendo (diamond-shaped), that radiates bilaterally to the carotids
- Handgrip
Decreases the intensity of the murmur.
- Valsalva and standing from squatting
Decreases or does not change the intensity of the murmur .
- Soft S2
- S4 is best heard at the apex.
- Early systolic ejection click



Echocardiography

Recommended primary test; to confirm diagnosis and determine AS severity

Chest x-ray

- Used to assess for pulmonary edema or other causes of dyspnea.
- Visible calcifications within the aortic valve may indicate more severe disease
- X-ray signs of LVH

Cardiac catheterization

- Accurate diagnostic test for the evaluation of aortic valve area
- Cardiac output
- Mean aortic pressure gradient

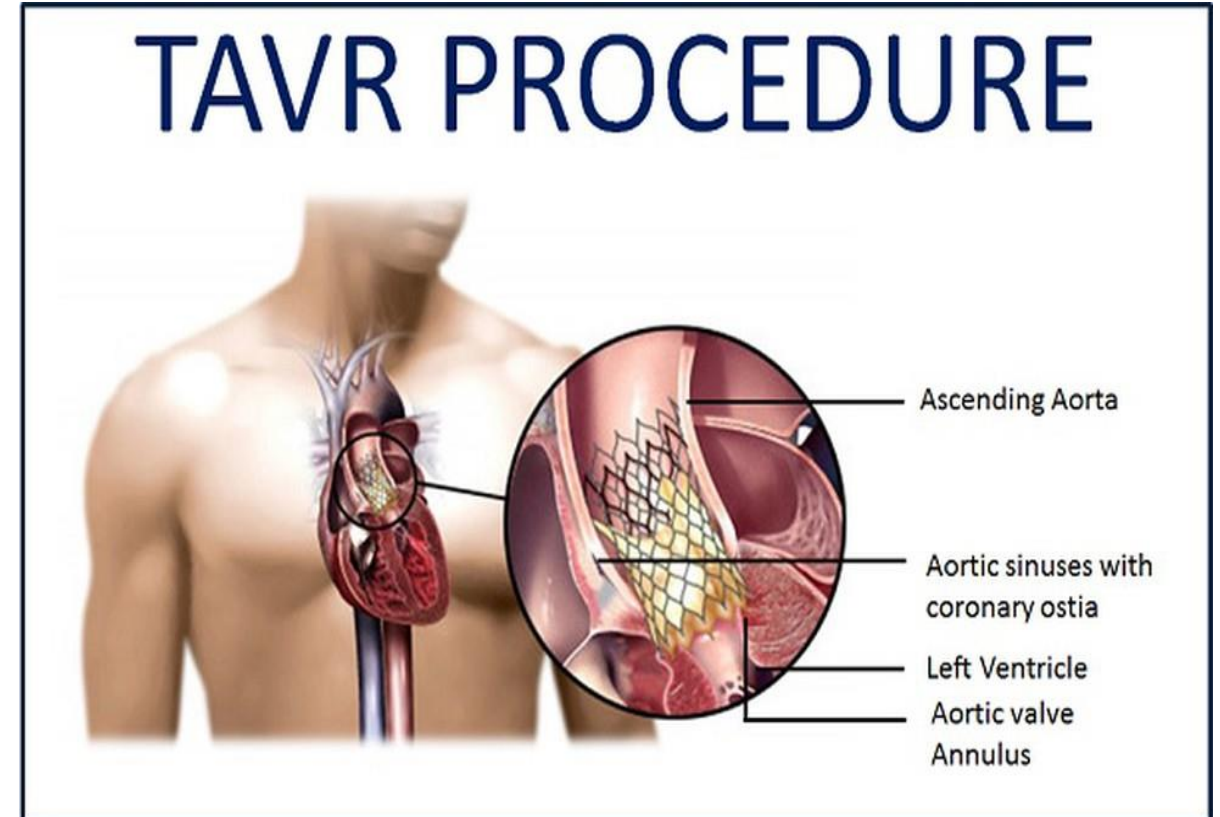
- **Aortic valve replacement (AVR) or repair**

Indications

- Symptomatic patients
- Asymptomatic patients with severe AS and:
 - Significantly ↓ LVEF
 - Undergoing cardiac surgery

Procedure

- Surgical AVR (SAVR)
- Transcatheter AVR (TAVR)
- Percutaneous balloon valvuloplasty





Valvular Heart Disease

Mitral regurgitation

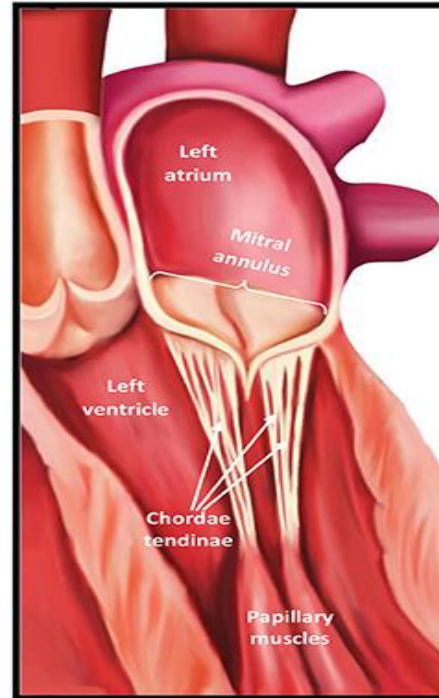
(MR)

Mitral regurgitation

Primary MR (organic):

Mitral regurgitation caused by direct involvement of the valve leaflets or chordae tendinae

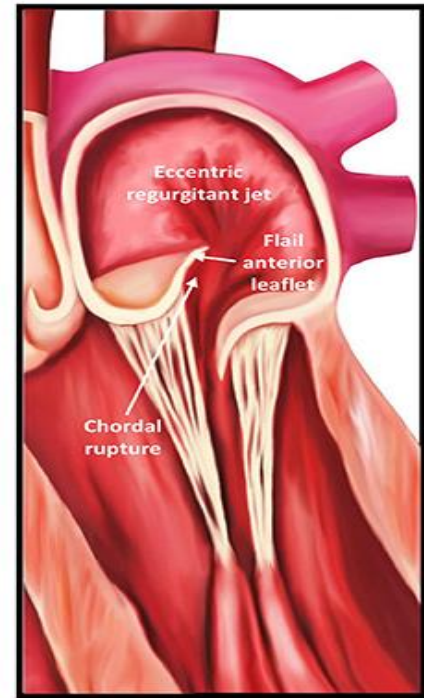
- Degenerative MV disease:
 - Mitral valve prolapse
 - Mitral annular calcification
 - Ruptured chordae tendinae
- Rheumatic fever
- Infective endocarditis



Normal mitral valve anatomy



Primary mitral regurgitation due to valve prolapse



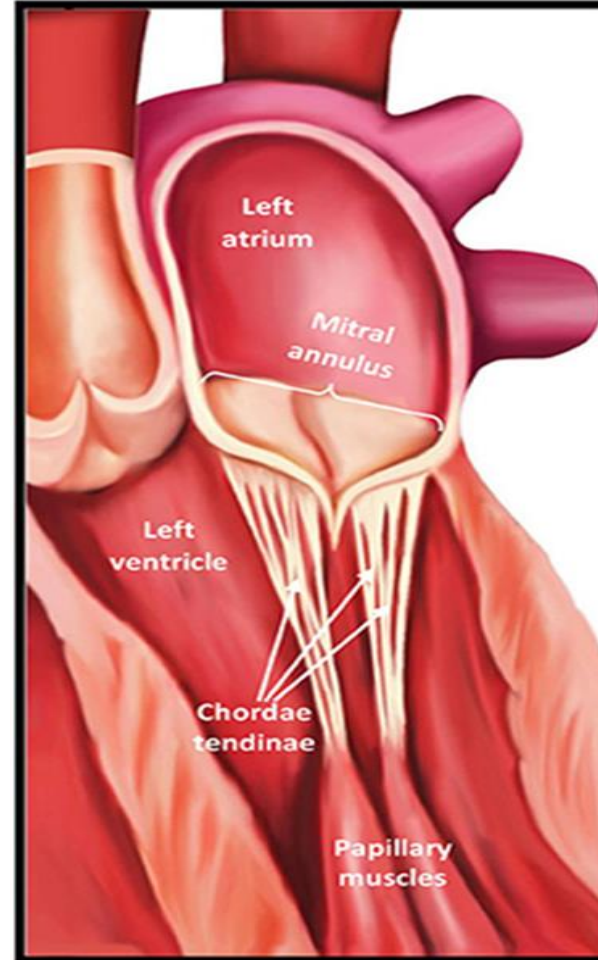
Primary mitral regurgitation due to flail leaflet

Mitral regurgitation

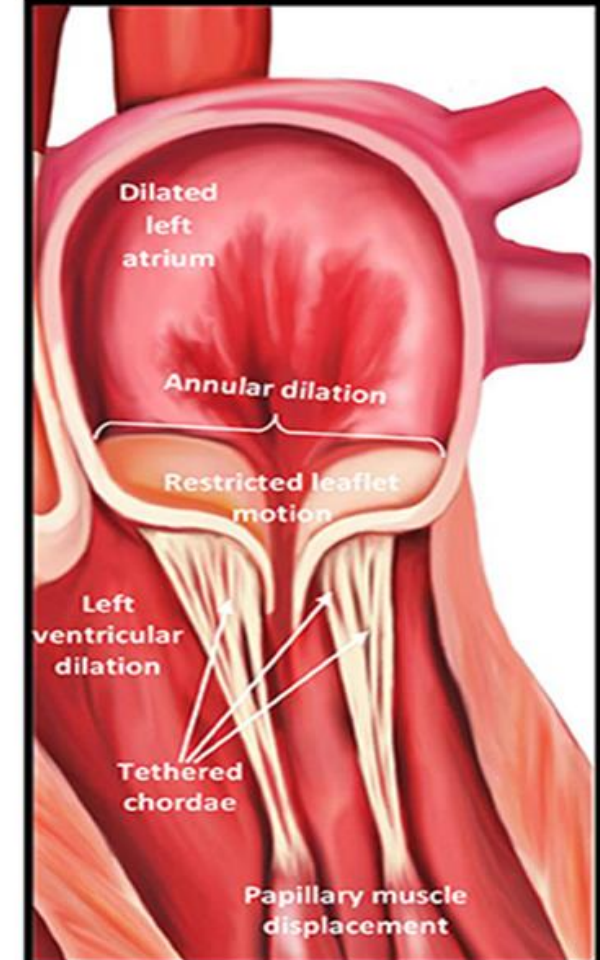
Secondary MR (functional):

caused by changes of the left ventricle that lead to valvular incompetence

- Coronary artery disease or prior myocardial infarction causing papillary muscle involvement
- Dilated cardiomyopathy



Normal mitral
valve anatomy



Functional mitral
regurgitation

Mitral regurgitation

Clinical features

Acute mitral regurgitation

- Symptoms of left-sided heart failure
- Signs and symptoms of pulmonary edema
- Cardiogenic shock

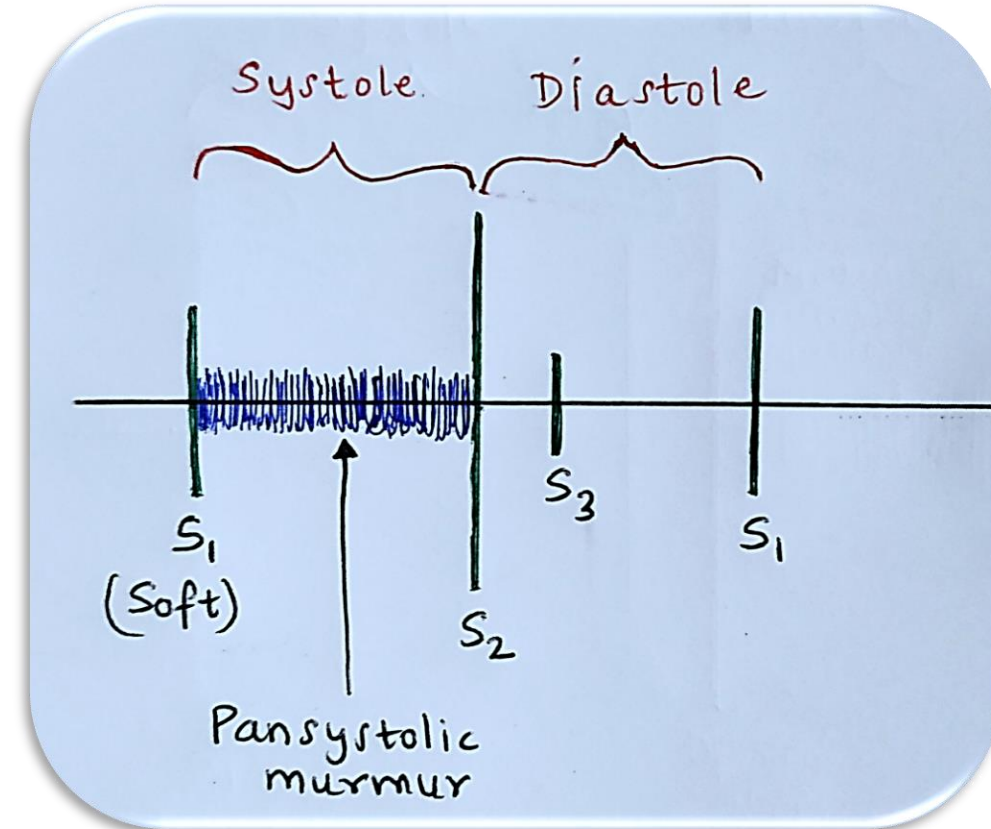
Chronic mitral regurgitation

- Dyspnea (including exertional dyspnea), dry cough
- Palpitations
- Symptoms of left-sided heart failure

Mitral regurgitation

Auscultation

- S3 heart sound in advanced stages of disease
- Holosystolic murmur (high-pitched, blowing)
- Radiates to the left axilla and heard best over the apex
- Intensity can be increased by:
 - Increasing preload (e.g., leg raise)
 - Increasing afterload (e.g., handgrip) due to increased regurgitation.



Mitral regurgitation - Diagnostics

Echocardiography

- Indications: to assess the valve apparatus, size and function of left ventricle and atrium, and grade the severity of MR
- TTE: modality of choice for the initial assessment of all patients with suspected valvular abnormality

ECG

- Chronic MR: ECG changes usually reflect cardiac remodeling.
- Left ventricular hypertrophy
- P mitrale

Chest x-ray

- Signs of pulmonary congestion
- LV enlargement
- LA enlargement

Mitral regurgitation - Treatment

Acute mitral regurgitation

All patients with acute primary MR should undergo urgent surgical repair or replacement.

While awaiting surgery, any symptoms of heart failure should be managed with medical therapy (e.g., diuretics, nitrates, antihypertensive drugs).

Indications of surgery

Acute primary MR: emergency/urgent surgery

Procedures

Valve repair: preferred option

Mitral regurgitation - Treatment

Chronic mitral regurgitation

- Surgical management
- Transcatheter mitral repair

Indications:

severe primary MR with any of the following:

- Symptomatic patients irrespective of LV systolic function
- Asymptomatic patients with LV systolic dysfunction; $LVEF \leq 60\%$





Valvular Heart Disease

Mitral Valve Prolapse

(MVP)

Mitral Valve Prolapse

Epidemiology

The most common cause of mitral regurgitation in developed countries

Etiology

- Mostly idiopathic
- Marfan syndrome,
- Ehlers-Danlos syndrome
- Acute rheumatic heart disease
- Infective endocarditis

Mitral Valve Prolapse

Pathophysiology Myxomatous degeneration

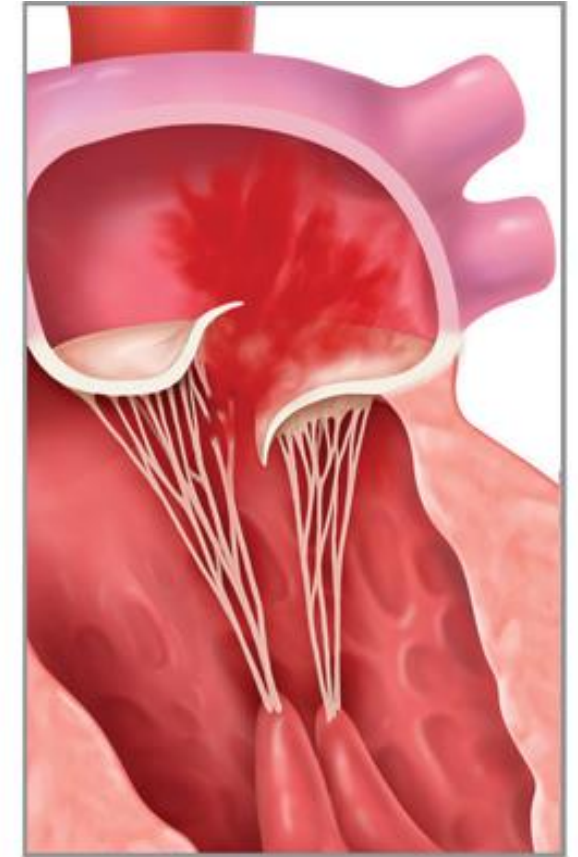
- MVP
- Flail MV leaflets



Normal Valve



Degenerative MR—
Prolapse



Degenerative MR—
Flail

Mitral valve prolapse

Clinical features

Most patients are asymptomatic

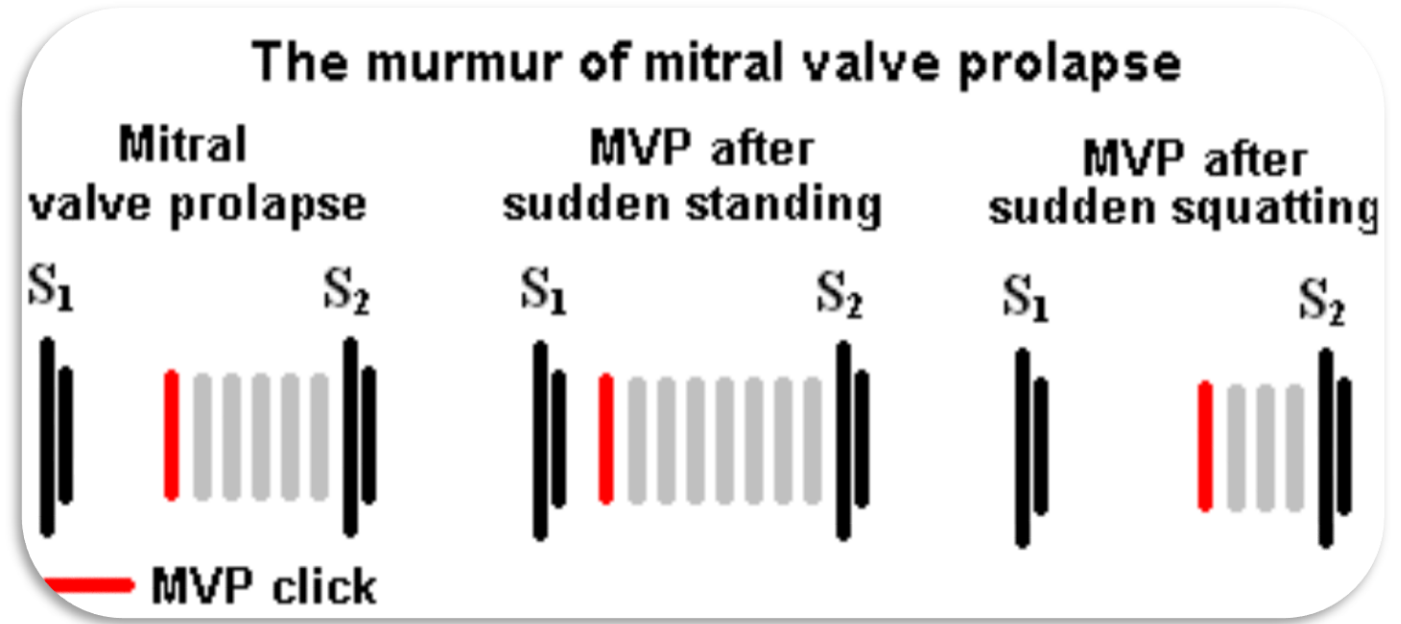
Auscultatory findings

Mitral valve prolapse click:

high-frequency, midsystolic click
best heard at the mitral region

High-frequency, mid-to-late systolic murmur

best heard at the mitral region and may radiate to the axilla
(**squatting diminishes the murmur**)



Mitral valve prolapse

Diagnostics

Transthoracic echocardiography (test of choice)

Treatment

No treatment is required in most cases.

Complications

Infective endocarditis



Valvular Heart Disease

Mitral Valve Stenosis

(MS)

Mitral Valve Stenosis

Etiology

Most commonly due to rheumatic fever

Clinical features

- Dyspnea
- Hoarseness
- Dysphagia

Later stages

- Symptoms of right heart failure
- Paroxysmal nocturnal dyspnea
- Orthopnea
- Hemoptysis

Malar flush vs. malar rash

Malar flush (mitral facies):

pinkish purple patches on cheeks, peripheral cyanosis in lips, tip of nose and cheek.



Malar Rash / Butterfly Rash

Erythematous flat or raised rash across the bridge of the nose and cheeks, which usually spares nasolabial folds.

Commonly seen in:

- Erysipelas
- Cellulitis
- Systemic lupus erythematosus
- Rosacea
- Pellagra
- Dermatomyositis

Precipitated by exposure to sunlight

How is it different to MALAR FLUSH?

The malar flush has a similar appearance, but it is a generalized reddening caused by a vasomotor reaction, whereas butterfly rash is a dermatitis composed of specific lesions (eg macules, papules).
Seen in - Mitral stenosis, Polycythaemia rubra vera



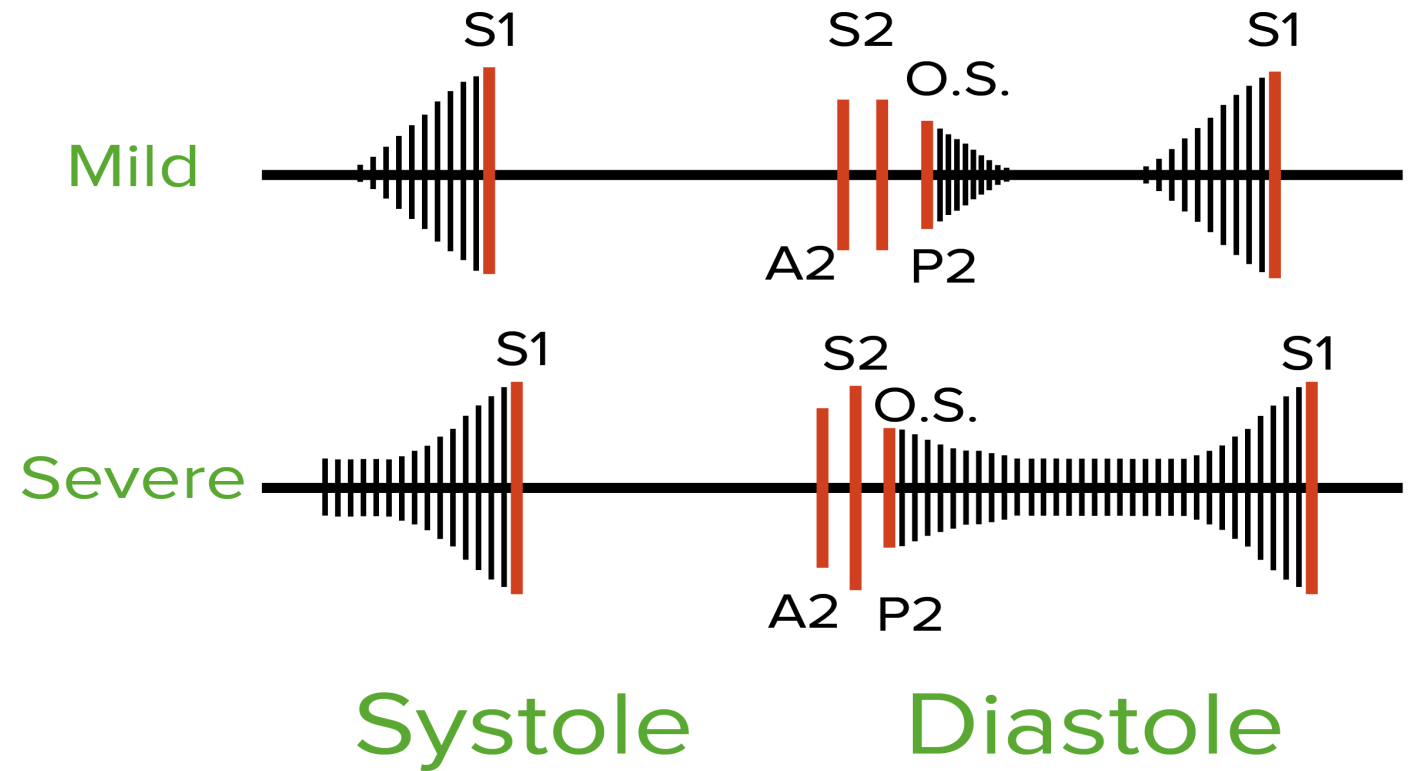
Mitral Valve Stenosis

Examination findings

Auscultation

- Diastolic rumbling murmur heard best at the apex
- Loud first heart sound
- Opening snap

Mitral stenosis



Mitral Valve Stenosis - Diagnostics

Transthoracic echocardiography (TTE)

TTE is the most important test for diagnosing and guiding the treatment of mitral stenosis.

- Reduced mitral valve area (MVA)
- Thickened, calcified leaflets with commissural fusion
- LA enlargement

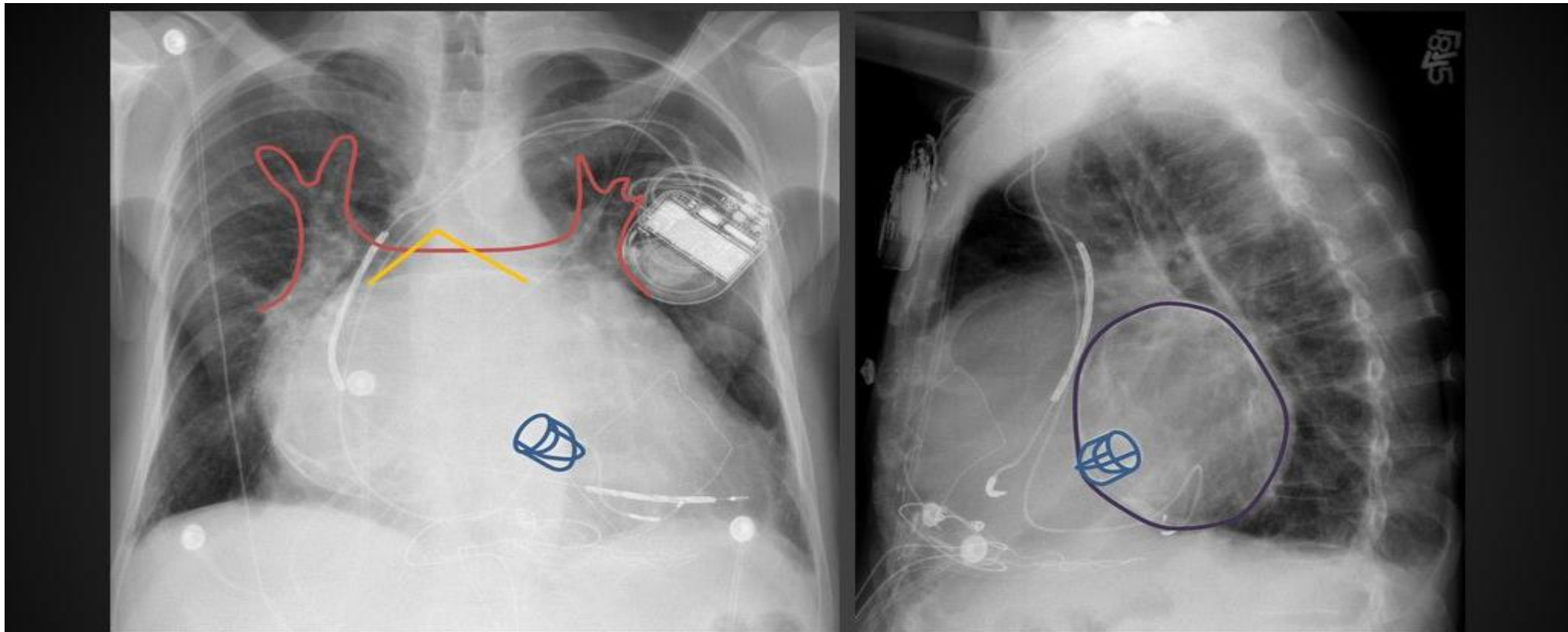
ECG

- Left atrial enlargement = P mitrale

Mitral Valve Stenosis - Diagnostics

Chest x-ray (PA and lateral views)

- Left atrial enlargement
- X-ray features of pulmonary congestion
- Enlarged left atrial appendage



Acute stabilization

Heart failure symptoms are managed using standard therapy, e.g., diuretics.

Interventional management

Procedures

Percutaneous mitral valve balloon commissurotomy (PMBC) is the preferred intervention in most patients with severe MS.

Surgery

Surgical interventions include open commissurotomy and mitral valve (mechanical or bioprosthetic) replacement.

Mitral Valve Stenosis - Complications

Atrial fibrillation
Pulmonary edema,
Pulmonary hypertension
Congestive heart failure

Tricuspid Regurge (TR)

Aetiology: TR is usually functional resulting from RVE → dilatation of the tricuspid ring.

Hemodynamic: During systole, part of the blood regurgitates from RV to RA causing:

↑ RA pressure → RA enlargement & systemic congestion. LCOP.

Clinical Picture: of RV failure

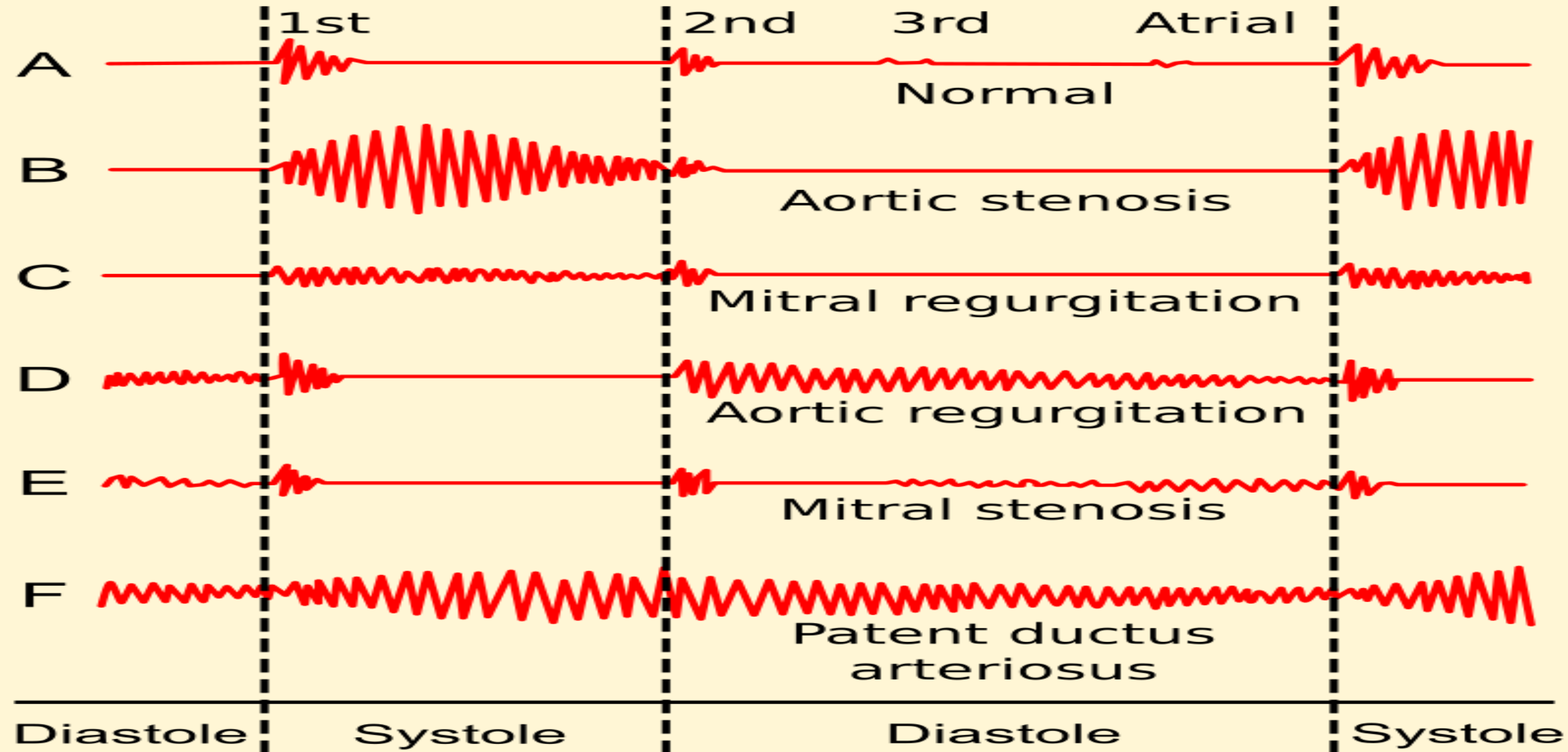
Symptoms: - Of the cause. - Systemic congestion. -LCOP

Signs of systemic congestion :

- Congested pulsating **neck vein** with systolic expansion.
- **Enlarged tender** pulsating **liver** with mild jaundice.
- **Ascites** before **oedema LL**.
- **Signs** of LCOP: cold hand, weak pulse, low SBP, peripheral cyanosis.

Auscultation: **Pansystolic murmur, Increased by inspiration.** Maximum **over tricuspid area & propagated to the apex but not to the axilla.**

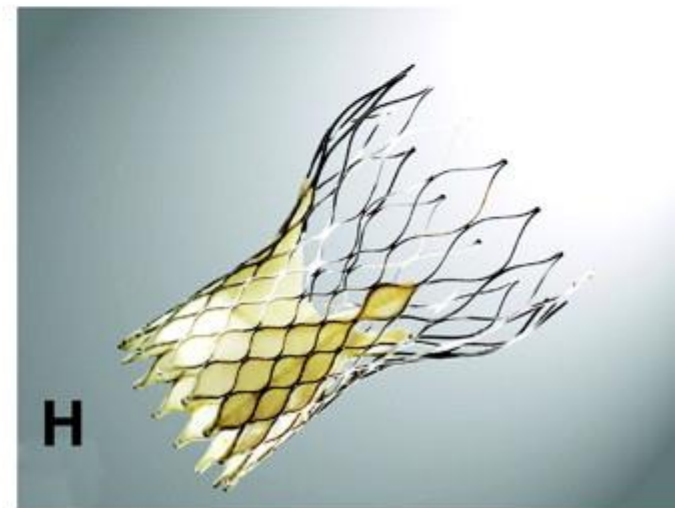
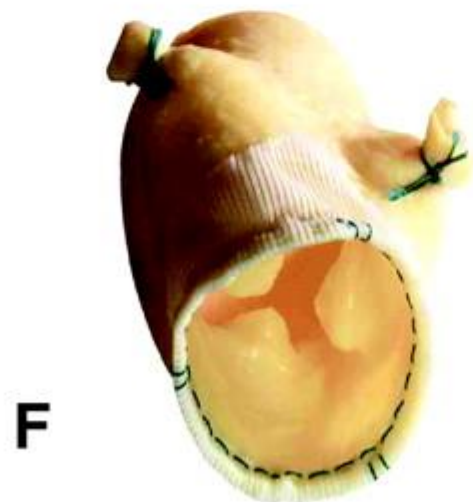
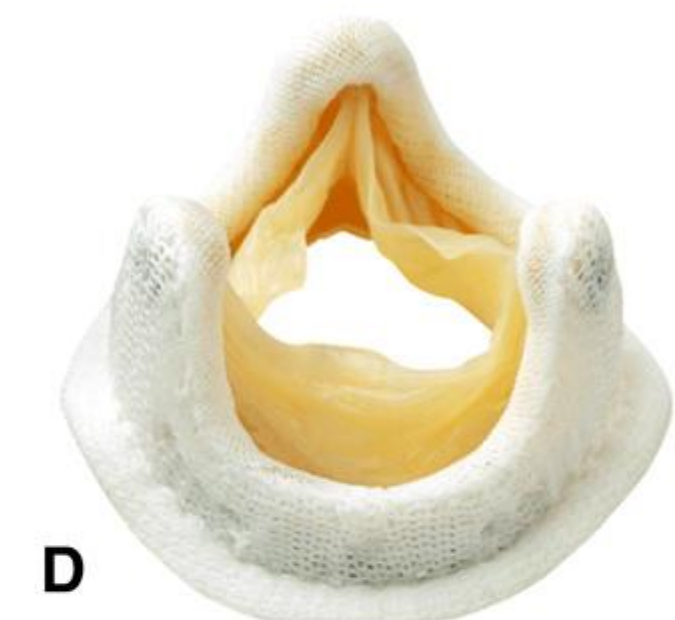
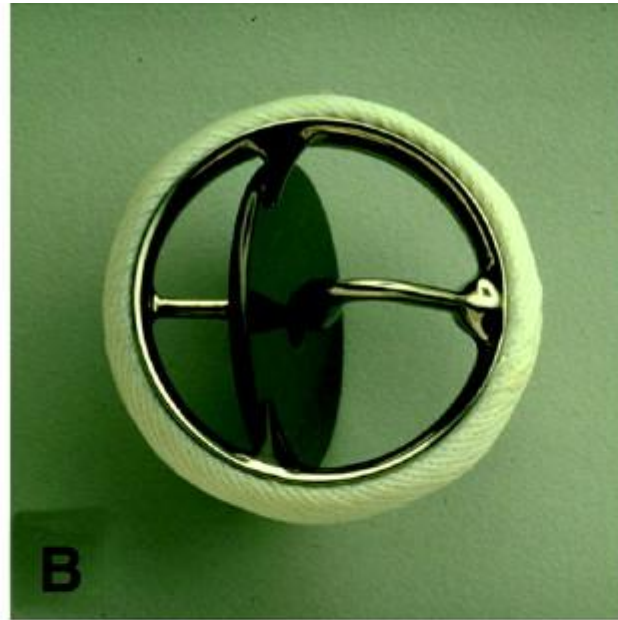
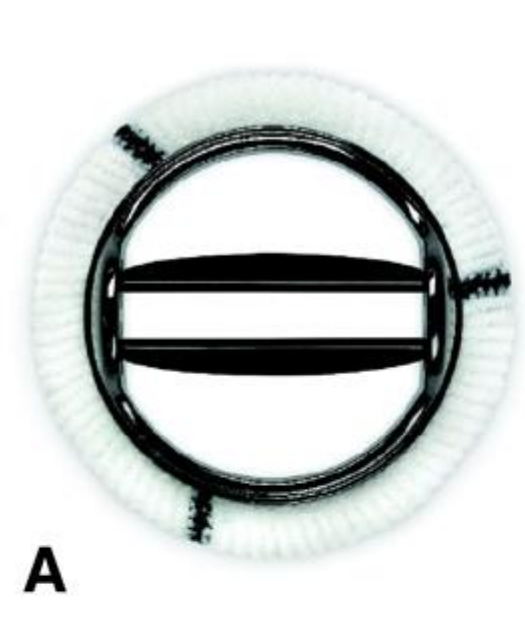
Cardiac Auscultation



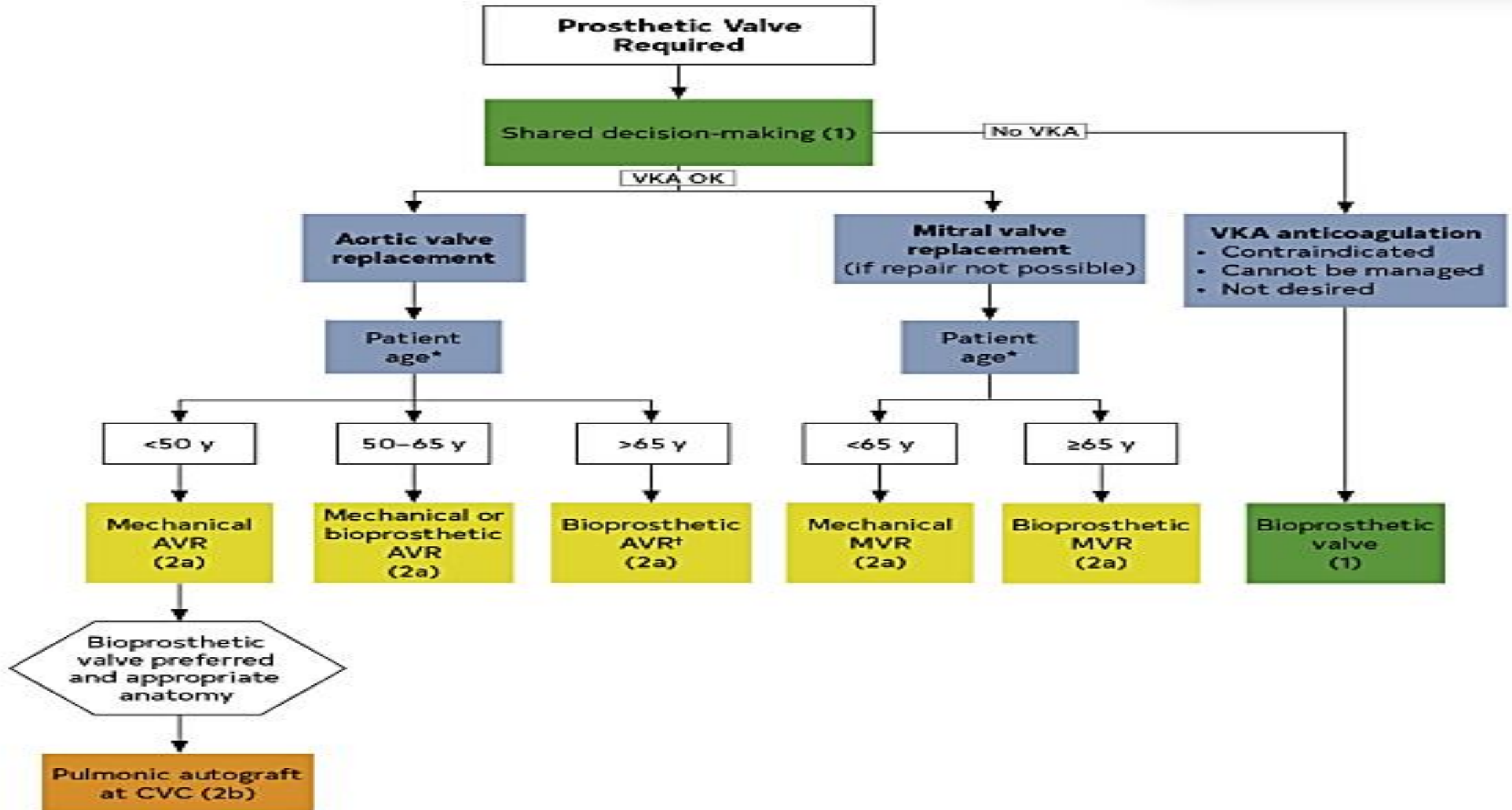


Prosthetic Valves

Types of prosthetic valves



Prosthetic Valves



Thromboembolic risk factors for prosthetic valves

- Older-generation valve
- AF
- Previous thromboembolism
- Hypercoagulable state
- LV systolic dysfunction

References & further reading

- 2020 ACC/AHA Guideline for the Management of Patients With Valvular Heart Disease: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. J Am Coll Cardiol 2020;Dec 17
- Zipes DP. Braunwald's Heart Disease. Mosby; 2018
- AMBOSS
- Harrison's Principles of Internal Medicine 20th edition ISBN-13: 978-1259644030

Thank You

Questions

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